ChatGPT

Executive Summary

Exam.net and Synap both offer secure online exam platforms with built-in proctoring, but their approaches differ markedly. **Exam.net** is browser-based and geared toward ease of use: students join exams via a simple exam key (no account needed)【9†L76-L80】. It provides *lockdown modes* (up to "high security" requiring a special secure browser or app) and server-side cheat detection, but it explicitly **does not record** exam video or audio【9†L83-L89】. Teachers can monitor student progress and receive flagged events (e.g. quiz window exits) in real time, and communicate via an integrated chat【7†L139-L147】【6†L62-L70】. In contrast, **Synap** (with its "Synoptic" proctoring) offers a multi-level proctoring model built into the exam system. Administrators choose an "install level" (no install, a Chrome extension, or a full desktop lockdown app) and a separate "monitoring level" (no recording, periodic snapshots, or full video/audio recording)【16†L175-L184】【16†L313-L321】. Synap's platform actively captures the candidate's screen, webcam and audio (as configured), and uses AI vision to flag anomalies (phones, second faces, eye gaze, etc.)【33†L474-L483】【33†L498-L504】. Synap also supports live human proctoring and provides rich post-exam review tools (event timelines, flagged-video clip playback, etc.)【16†L350-L359】【33†L474-L483】.

In sum, Synap offers **richer proctoring features** (integrated camera/audio recording, AI analysis, flexible lockdown) at the cost of greater technical complexity, whereas Exam.net emphasizes **simplicity and privacy** (no video recording, browser-only setup)【9†L83-L89】【16†L313-L321】. Both systems meet strict privacy/compliance standards (GDPR, FERPA, etc.)【9†L83-L89】【30†L49-L58】, but Exam.net's design favors minimal user friction, while Synap's UX is more feature-dense. Based on these findings, **EPS Phase 4** should implement mandatory lockdown and monitoring (like exam windows lock) and teacher-review tools (must-have), while considering optional AI vision features. Recommendations below prioritize key features (lockdown, logging, teacher UI) and outline technical design, a migration checklist, and a phased implementation plan (see MerMAID gantt).

Proctoring Feature Inventory and Parity

Feature	Exam.net	Synap (Synoptic)	Notes (implementation)	Confidence (source)
Student access	**Exam Key only** – no student accounts; students join via key【9†L76-L80】.	**Candidate accounts/portal** – students log in at their provider's Synap portal (email/password); "account lookup" via email is available【23†L72-L75】【21†L132-L139】.	Exam.net uses anonymous exam keys. Synap requires user accounts per provider, supporting SSO (SAML/OIDC)【30†L178-L181】.	High (Exam.net doc; Synap docs)

Feature	Exam.net	Synap (Synoptic)	Notes (implementation)	Confidence (source)
Teacher/Admin login	Yes – teachers have accounts and sign in (username/ password).	Yes – exam creators and admins log in via Synap portal (supports SSO) 【30†L178-L181】.	Both systems have role-based logins; Synap's "Trust Center" notes SAML/AD integration and MFA【30†L178-L181】.	High (Exam.net & Synap docs)
Registration (Teacher)	Not public – teachers sign up or get invited by institution (unspecified).	Admins/ instructors are invited by the organization's portal; Synap has role-based user management.	(Exam.net doesn't detail self-registration publicly.) Synap supports role-based access (admin creates teachers)【30†L178-L181】.	Medium (Exam.net unverified; Synap docs)
Device readiness/ system check	**None publicly** – assumes any modern browser.	**Pre-exam check** – portal runs a system check on browser, camera, connection 【23†L72-L75】.	Exam.net does not describe a device self-test. Synap guides candidates through a setup wizard and system check【23†L72-L75】.	High (Exam.net unverified; Synap guide)
Delivery mode	**Browser-based** (any modern browser). High-security mode uses *native or secure-browser app* (Safe Exam Browser, Exam.net App or kiosk)【6†L49-L51】.	**Browser, Extension or App** – No Install (just web), or a Chrome extension, or a Desktop App for full lockdown 【16†L199-L207】【23†L93-L102】.	Exam.net defaults to pure web; high mode requires specialized software. Synap explicitly offers optional extension and desktop app with each exam 【16†L225-L233】【23†L93-L102】.	High (Exam.net and Synap docs)
Full-screen lock	Yes – exam runs full-screen; in highest mode student *cannot leave* fullscreen except via teacher permission 【6†L49-L51】【9†L66-L72】.	Yes – in Desktop App mode, exam is locked full-screen. Extension mode can **detect** (flag) any exit from fullscreen 【16†L257-L264】.	Exam.net enforces fullscreen lock on highest security. Synap's Desktop app *prevents* minimization/ screenswitch; extension will log breaches if configured【16†L257-L264】.	High (Exam.net [6][9]; Synap [16])

Feature	Exam.net	Synap (Synoptic)	Notes (implementation)	Confidence (source)
Browser focus/ tab monitoring	Partial – not explicitly documented in UI. Likely "leave exam" attempts are flagged or blocked.	Yes – monitors tab focus. In No-Install mode, switching tabs is flagged in real time【16†L199-L207】.	Synap's "No Install" setup flags extra tabs/ losing focus. Exam.net's docs don't mention tab focus specifically (assumed by leaving fullscreen) – treat as unverified.	Low (Exam.net unverified; Synap [16])
New-tab (URL) detection	Not documented (likely only via fullscreen lock).	Yes – extension flags opening new tabs; Desktop app *blocks* all other apps/browser windows【16†L257-L264】.	Synap's Browser Extension can detect/ flag new tabs. Exam.net may rely on preventing web navigation (no mention in docs).	Medium (Synap [16]; Exam.net no doc)
Website blocking	High mode – blocks external sites by lockdown; in lower modes students could browse but may require teacher permission to resume exam【9†L66-L72】.	Extension mode: detects navigation to outside sites; Desktop app: blocks other websites entirely【16†L257-L264】.	Exam.net high-security blocks all other browsing by locking the test-browser【9†L66-L72】. Synap's extension will detect breaches if config set; app prevents it outright【16†L257-L264】.	High (Exam.net [9]; Synap [16])
Application blocking	High mode (app/ SEB) can lock device and prevent switching apps【6†L49-L51】.	Yes – Desktop app blocks launching other applications. Extension **cannot** block apps (only browser)【16†L257-L264】.	Exam.net's high-security modes (native apps) lock down device【6†L49-L51】. Synap's desktop client enforces app lockdown; browser extension has no control beyond the browser.【16†L257-L264】.	High (Exam.net [6]; Synap [16])

Feature	Exam.net	Synap (Synoptic)	Notes (implementation)	Confidence (source)
Prevent use of AI tools	Not mentioned.	Yes – Synap explicitly **blocks AI tools** on extension and app (highest security) 【16†L225-L233】.	Synap notes "AI tools blocked" in extension/ desktop modes 【16†L225-L233】. Exam.net does not cite blocking AI (so unverified if possible).	High (Synap [16]; Exam.net no info)
Prevent screenshots	Not documented (likely not done in browser mode).	Desktop app: prevents screenshots. Browser extension: cannot block OS screenshots 【16†L257-L264】.	Exam.net makes no mention of screenshot prevention. Synap's Desktop App explicitly stops screenshots; extension cannot (and only logs events) 【16†L257-L264】.	High (Synap [16]; Exam.net no info)
Screen recording (video)	**No** – explicitly *no automated recording* of exam session 【9†L83-L89】.	Optional – in "Video & Audio" mode, continuously records screen 【32†L1-L4】; "Snapshots" mode takes periodic images【16†L311-L319】; or off.	Exam.net does *not* record student screens 【9†L83-L89】. Synap can record full-screen video in high mode 【32†L1-L4】 or take snapshots every ~5s 【16†L311-L319】.	High (Exam.net [9]; Synap [16][32])
Webcam recording/ video	**No** – exam.net does not capture video【9†L83-L89】.	Yes – in higher monitoring levels, Synap captures webcam video/ snapshots 【19†L25-L32】 【32†L1-L4】.	Exam.net provides *no* webcam proctoring. Synap can record webcam: either periodic stills or full video 【19†L25-L32】 【32†L1-L4】 when enabled.	High (Exam.net [9]; Synap [19][32])
Audio recording	**No** – not mentioned, so no audio capture.	Yes – Synap can optionally record microphone audio along with video 【32†L1-L4】.	Exam.net has no audio feature. Synap's full recording mode includes audio【32†L1-L4】.	Medium (Synap [32]; Exam.net no info)

Feature	Exam.net	Synap (Synoptic)	Notes (implementation)	Confidence (source)
Periodic snapshots	**Unverified** (no mention).	Yes – "Recreation & Snapshots" mode takes webcam/screen images every ~5s 【16†L311-L319】.	Synap offers snapshots for moderate-stakes (audit trail without continuous video) 【16†L311-L319】. Exam.net's docs do not mention auto-screenshots.	High (Synap [16]; Exam.net no info)
AI anomaly detection	**None** – Exam.net relies on heuristic/server checks, not CV.	Yes – Post-exam AI review flags objects (phones, books, multiple people)【33†L474-L483】【33†L498-L504】.	Synap's AI "Segmentation" flags anomalies (mobile phone, second person, leaving frame) 【33†L498-L504】 and highlights them. Exam.net's monitoring is not AI-based.	High (Synap [33]; Exam.net none)
Live human proctoring	No – no built-in live view (teacher sees status and chat only).	Yes – with full recording, human proctors can watch candidate's screen/webcam live, send warnings 【16†L350-L359】.	Synap allows instructors to join live view (included with monitoring) and intervene with messages【16†L350-L359】. Exam.net offers no real-time camera feed.	High (Synap [16]; Exam.net no feature)
Teacher chat	Yes – students can chat privately with the teacher during exam 【7†L139-L147】.	Not mentioned – Synap's focus is exam content; no built-in exam chat noted.	Exam.net explicitly has a chat function for help 【7†L139-L147】. Synap's docs do not list a chat feature for candidate–proctor in exam.	High (Exam.net [7]; Synap none)
Answer history review	Yes – teachers can review each student's answer history/timing after exam (in Monitoring view) 【41†L1-L4】.	Unverified – not specified in publicly available docs.	Exam.net lets teachers click through answer history per student 【41†L1-L4】. Synap likely logs attempts but no doc found on per-question history.	High (Exam.net [41]; Synap no doc)

Feature	Exam.net	Synap (Synoptic)	Notes (implementation)	Confidence (source)
Plagiarism check	Yes – supports Turnitin/ Ouriginal integration (requires license) 【8†L64-L68】.	Not built-in – synap focuses on exam proctoring, no mention of plagiarism checks.	Exam.net can send student text to plagiarism tools 【8†L64-L68】. Synap does not mention plagiarism integration in its proctoring materials.	High (Exam.net [8]; Synap none)
Answer anonymization	Yes – option to anonymize student names during marking 【7†L117-L125】.	Not mentioned – likely outside proctoring scope.	Exam.net can hide identities for grading fairness 【7†L117-L125】. Synap docs focus on proctoring; anonymity is not addressed.	High (Exam.net [7]; Synap no mention)
Data encryption & compliance	Complies with GDPR, FERPA, COPPA, etc 【9†L83-L89】 【29†L36-L44】. Data not sold.	ISO 27001 certified, GDPR/ FERPA compliant 【30†L49-L58】. Data is owner-controlled, encrypted 【30†L91-L100】 【30†L115-L123】.	Both platforms emphasize privacy. Exam.net notes compliance and that it won't share student data 【29†L36-L44】. Synap has formal certs and encryption 【30†L49-L58】 【30†L115-L123】.	High (Exam.net [29]; Synap [30])

Key: "Unverified" indicates a feature is not documented in available sources. All stated capabilities are drawn from the official product pages and support documents of Exam.net and Synap.

Authentication & Registration

Exam.net uses a lightweight authentication model. **Teachers/administrators** have accounts and sign in to the Exam.net platform, but **students do not create accounts** – they simply enter an exam key to start an exam 【9†L76-L80】. (This means no student credentials are stored by Exam.net.) In contrast, Synap requires each candidate to have an account on the organization's Synap portal. Students log in (or recover access via "account lookup") at a custom subdomain (e.g. `school.synap.ac`) 【23†L72-L75】 【21†L132-L139】. Synap supports enterprise SSO (SAML/OpenID) for providers 【30†L178-L181】, while Exam.net's site suggests standard password logins. The registration flow will likely be very different: EPS can borrow the idea of "exam keys" for student access (no account needed) to simplify workflow, as Exam.net does, or implement accounts as Synap does (with added user management).

Device Readiness and System Checks

Synap explicitly provides pre-exam system checks. Before a Synoptic exam, candidates are guided through a setup wizard that verifies the browser, camera, and internet connection【23†L72-L75】. The portal also explains download requirements (Chromium extension or desktop app if needed)【23†L95-L104】【23†L102-L110】. Exam.net's documentation makes no mention of a readiness checker. It simply assumes a "modern web browser," except that its high-security mode requires installing a secure app or browser extension【6†L49-L51】. Thus, EPS should include a **device compatibility page** (as a must-have) – at minimum checking browser version and webcam/mic access – inspired by Synap's approach【23†L72-L75】.

Browser Monitoring & Lockdown Modes

Both platforms offer exam lockdown, but in different ways. Exam.net provides multiple security **levels**. In low-security mode, any modern browser may be used; in high security, students must use the Exam.net app or Safe Exam Browser, which fully *locks down the device* (disabling other apps on Chromebooks, iPads, PCs)【6†L49-L51】. Even in normal modes, Exam.net forces fullscreen and prevents exiting without permission【9†L66-L72】. Synap's "install levels" achieve similar goals:

- **No Install (Browser)** – minimal setup; the exam runs in the browser with Synoptic controls active. Tabs and focus are monitored (switching away can be flagged)【16†L199-L207】.
- **Browser Extension** – candidates install a Chrome extension. This blocks access to outside websites and AI tools, and detects violations (fullscreen exit, new tabs)【16†L225-L233】【16†L257-L264】.
- **Desktop App** – a full installer for Windows/Mac that enforces total lockdown (no external apps/sites, no screenshots)【16†L241-L249】【16†L257-L264】.

In practice, both systems prevent most forms of digital cheating. Exam.net's highest mode and Synap's Desktop App fully *prevent* leaving the exam, whereas Synap's browser modes generally *detect* and log attempted breaches【9†L66-L72】【16†L257-L264】. EPS should likewise implement a graded lockdown: at a minimum, force fullscreen and capture "focus lost" events (must-have). A Chrome extension or similar could optionally give stronger enforcement (should-have, but device permissions may be a barrier). Both platforms emphasize that their security is "integrated" and configurable by exam settings【9†L66-L72】【33†L474-L483】.

AI and Computer-Vision Proctoring

Exam.net does **not** use AI computer-vision. Its proctoring is based on browser control and server-side cheat algorithms【9†L83-L89】. Synap, however, actively uses AI vision. Synoptic captures webcam video and applies ML models to flag anomalies after the exam【33†L474-L483】. For example, its AI will automatically detect and timestamp events like a mobile phone in view, a second person's face, or the candidate leaving camera frame【33†L498-L504】. Importantly, Synap's AI only "points reviewers to what matters" – it never penalizes automatically ("flags, never verdicts"【33†L488-L496】). These flagged moments appear in the evidence review for human decision.

EPS Phase 4 should consider incorporating computer-vision **assistance**: for instance, use MediaPipe or TensorFlow.js face-mesh models (as Synap likely does) to detect multiple faces or off-screen moments. However, this is complex (object detection, head-pose, eye gaze) and can risk false positives. We categorize

it as *optional/advanced*: valuable for high-stakes exams but not mandatory. If included, it must be transparent and reviewer-driven, following Synap's "fair by design" ethos (AI as aide, not authority) 【33†L494-L502】【33†L553-L562】.

Recording and Evidence Handling

A key distinction is recording. **Exam.net explicitly does not record video/audio** of remote exams【9†L83-L89】; it relies on preventive controls and logs. Teachers get reports of "suspicious events" from server-side analysis (e.g. VM or remote-desktop use)【6†L60-L69】 but no video stream. Synap, on the other hand, can record everything. In its top monitoring level, Synap "continuously record[s] the candidate's screen, webcam and audio"【32†L1-L4】, generating a full audit trail. In mid-level mode, it instead takes periodic snapshots (~one every 5 seconds) of screen and camera, plus a reconstructed timeline【16†L311-L319】. In either case, all recordings are uploaded and accessible post-exam.

Thus **evidence management** differs: Exam.net gives teachers data points and logs (browser-focus events, answer timestamps, answer history)【41†L1-L4】 but no media. Synap offers video clips and images. For EPS, a baseline requirement is detailed logging (timestamped logs of exam actions). A recommended enhancement (should-have) is screen/webcam snapshots: capturing an image every few seconds (as Synap does)【16†L311-L319】. Full video recording is a major feature (optional) and requires storage. If implemented, it must be secured (encrypted in transit/rest【33†L552-L561】) and clearly communicated to users. Synap's candidate instructions emphasize that recordings are protected and decisions are made by instructors【23†L116-L120】 – a privacy model EPS should emulate.

Teacher Monitoring & Review Workflows

Exam.net provides a simple teacher dashboard. In the **Monitoring view**, instructors see who has started the exam, progress bars, and who has submitted【7†L139-L147】. Teachers can click on any student to see their answer history over time【41†L1-L4】, chat with them, and review any logged "cheat events" flagged by the system【6†L60-L69】. There is no live video feed.

Synap's instructor interface is richer: it shows real-time scores of attempts (with flags) and a detailed review page for each exam session【33†L468-L476】. Each session's review screen displays paired screen and webcam recordings (thumbnail timeline) and a list of event logs/comments【16†L350-L359】【33†L468-L476】. In "Live Proctoring" mode, an invigilator can watch the live webcam/screen feed and issue warnings 【16†L350-L359】. Afterward, Synap's AI highlights key moments (camera icons on the timeline)【33†L474-L483】.

Overall, teacher workflows are analogous: teachers must be able to **monitor during the exam** (class progress and chat) and **investigate after** (logs, events, or recordings). EPS should implement an easy-to-scan dashboard (must-have): e.g. a table of live sessions with status, plus a per-student timeline of events. Drawing inspiration from Synap's "Review" UI, include jump-to-timestamp for any flagged event (should-have). Tools like filtering by "screenshots" or "webcam frames" (as Synap offers) enhance scalability for long exams【33†L414-L423】.

Admin Controls

Both platforms support admin-level controls, though Exam.net's public docs focus on teacher roles. Synap explicitly supports role-based access (detailed in its Trust Center), with MFA, audit logs, and data ownership guarantees【30†L176-L184】. We infer EPS should include:

- **Role-based permissions**: separate teacher and administrator roles (must-have).
- **Data retention settings**: Synap allows configurable retention【33†L548-L556】; EPS should too (e.g. delete recordings after X days – should-have).
- **Compliance logging**: audit trails of user/admin actions (Synap's ISO controls suggest this【30†L174-L183】).

UX and Visual Design Patterns

Exam.net's public pages have a clean, minimal style: bright white backgrounds with orange accents, simple typography and flat UI elements. (For example, the Teacher dashboard uses clear progress bars and a straightforward grid【7†L139-L147】.) Forms (sign-in) are minimal and focused. Synap's site is more corporate: it uses a blue/grey/white palette and larger illustrations. Both emphasize clarity and responsive layouts.

For EPS, we recommend **minimalist, high-contrast design** with a warm accent color (e.g. orange or blue) for CTAs and status indicators. On the **Home page**, include multiple informative sections: platform overview, security badges (GDPR/ISO), and proctoring features, akin to Synap's structured sections【15†L49-L57】【30†L49-L58】. The **unified Auth page** should have a simple role switcher (Student vs Teacher/Admin), similar to Synap's portal login (Synap suggests different exam portals per organization【21†L65-L73】). A single-page sign-in/registration card with bold labels and subtle borders fits the minimal style.

Exam.net's **exam room UI** is fully white with tool panels (chat, question nav) flush to the sides【7†L139-L147】. We should adopt a similar layout: main central area for question content, with collapsed sidebars for chat and tools. Questions are presented one at a time (as in Synap's screenshot【15†L59-L68】). UI buttons (Submit, Next, Zoom, etc.) should be clearly distinguishable in the accent color. For **teacher dashboards**, use card-based tables with consistent row highlighting. Follow modern SaaS patterns: collapsible side menus, icons for alerts (e.g. a red flag for breaches), and clear "Approve/Reject" buttons in the evidence review panel. In summary, blend Exam.net's white/clean aesthetic with Synap's structured content approach, maintaining generous whitespace and readability.

Security, Privacy & Consent Practices

Both products prioritize data protection. Exam.net explicitly states it complies with GDPR, FERPA, COPPA and that *"we never disclose personal information about you or your students to any third party"*【29†L36-L44】. It emphasizes that *no student data is used for secondary purposes*. Crucially, Exam.net **does not record** students' video or audio【9†L83-L89】, minimizing privacy concerns. Its Privacy Center notes that any student info is collected only as needed to run exams, and students do not even need accounts【29†L36-L44】.

Synap likewise touts ISO 27001 certification and global privacy compliance【30†L49-L58】. Importantly, Synap pledges *"Your data stays yours — never sold, shared, or used to train AI"*【30†L49-L58】 and explicitly states recordings are encrypted in transit/rest【23†L116-L120】. Synap lets clients choose data residency (EU/US) and configure retention【33†L548-L556】. On Synoptic exams, candidates are informed of what's being captured, and review decisions are made by the institution【23†L116-L120】. Both platforms use secured, audited infrastructure (Synap on AWS with 99.9% SLA【30†L65-L73】).

In practice, EPS should mirror these practices: obtain clear consent (especially if using cameras), minimize data collection (e.g. skip recording if not needed), and encrypt all stored evidence. Publish a privacy statement covering recording (like Synap's guide【23†L116-L120】) and retention policies (time-limited access links【33†L548-L556】).

Technical Constraints & Considerations

Key constraints for browser-based proctoring include the available web APIs and performance limits. Capturing video/audio uses **WebRTC / getUserMedia** and **MediaRecorder** in the browser. Continuous recording and computer-vision (face/pose) can strain CPU/GPU, especially on lower-end devices. Tools like MediaPipe FaceMesh or TensorFlow.js (both open-source) could run client-side for eye/head tracking, but real-time object detection (like phone vs textbook) may require powerful models. Indeed, Synap notes it uses open-source models on its own infra (no third-party ML services)【33†L553-L562】.

A browser extension (as Synap uses) can access more APIs (e.g. tab events) but only on Chrome; others must rely on Web Extensions API. A native desktop app (Electron or similar) can enforce deeper lockdown (blocking OS-level screenshots, apps), but adds installation overhead. WebRTC security prompts must be handled carefully. Licensing can also be an issue: exam solutions must avoid using non-free third-party services for face recognition.

EPS should therefore plan for: browser window and fullscreen API, Page Visibility API for focus changes; Screen Capture API or tab capture for recording; TensorFlow.js or MediaPipe (Apache 2.0) for ML; secure storage (e.g. AWS S3 with AES-256). All external libraries should be open-source or self-hosted, per privacy needs【33†L553-L562】.

Prioritized Recommendations for Phase 4

1. **Must-Have:** *Exam Lockdown & Monitoring*. Implement fullscreen exam mode and detect any attempt to switch away or open other tabs. Log these events in real time for teacher alerts【9†L66-L72】【16†L257-L264】. (Complexity: moderate; relies on standard browser APIs; low risk.)
2. **Must-Have:** *Teacher Dashboard & Logs*. Build a live dashboard showing each student's status (started, time, flags) and chat. Include post-exam review with timestamped event logs【7†L139-L147】【33†L468-L476】. (Complexity: moderate; standard UI work; low risk.)
3. **Must-Have:** *Privacy Safeguards*. Ensure all data (including any recordings) are encrypted and self-owned (no external sharing)【29†L36-L44】【30†L49-L58】. Explicitly disclose to candidates what is captured. (Complexity: low; mostly policy and secure engineering; low risk.)
4. **Should-Have:** *Video Snapshot Capture*. Periodically capture screenshot images of the exam screen and/or webcam for audit trails. Store these efficiently (e.g. one JPEG per 5–10 seconds). (Complexity: moderate to high; needs MediaRecorder/Canvas and storage; mid risk.)

5. **Should-Have:** *MediaPipe Face/Gaze Detection*. Integrate a face-detection model (e.g. FaceMesh) to detect if the user leaves view or looks away. Flag such events in logs【33†L498-L504】. (Complexity: high; performance-sensitive, but open-source code available; moderate risk of false positives.)
6. **Optional:** *Full Video Recording*. Offer continuous recording of screen/webcam with user permission, for very high-stakes exams【32†L1-L4】. (Complexity: very high storage and bandwidth; significant risk/overhead.)
7. **Optional:** *Live Proctoring*. Enable human proctors to join live exam sessions (with dashboard chat/warning). (Complexity: high (streaming); might use WebSocket; moderate risk.)
8. **Optional:** *AI Anomaly Detection*. Use TensorFlow.js to automatically flag common infractions (e.g. detect phone, cheating posture) as Synap does【33†L498-L504】. (Complexity: very high; requires ML infrastructure; high risk of inaccuracy.)

Each recommendation notes its estimated complexity and risk. Must-have items should be tackled first (Phases 4a/b), while “should” and “optional” features can be phased after core functionality.

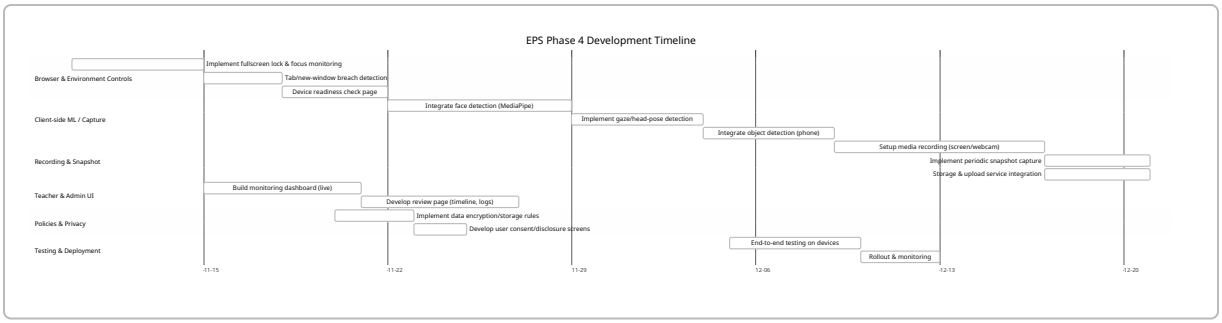
Migration Checklist

To integrate proctoring into the existing Django EPS, prepare the following:

- **New Models:**
- `ProctoringSetting` (per-exam config: lockdown level, monitoring level, use webcam, etc).
- `ExamSession` (tracks an individual exam attempt, linking student, exam, start/end times).
- `ProctorEventLog` (timestamped events: focus-lost, tab-change, snapshot file refs, AI flags).
- `ProctorRecording` (file references for video/snapshots and metadata).
- **Endpoints & Frontend:**
- Endpoint to start/end exam sessions (begin recording stream).
- WebSocket or AJAX hooks to send events (e.g. `focus`, `violation`, or snapshot data) to server in real time.
- Upload URLs for video/snapshots (e.g. S3 pre-signed).
- Frontend scripts (JavaScript) to handle fullscreen lock, getUserMedia, and ML detection if used.
- **Storage:**
- Set up object storage (cloud or on-prem) for large media.
- Database migrations for new models (with constraints/indices).
- **Permissions:**
- Ensure only teachers can access event logs and media for their students.
- Protect recording URLs behind authenticated links (time-limited tokens).
- Students cannot view logs or recordings (only submit to EPS).
- **Testing:**
- Unit tests for new views and models.
- Integration tests simulating exam flows (e.g. a fake exam session, logging events, teacher review).
- End-to-end QA on different devices and browsers (the candidate readiness check should be tested on Chrome, Firefox, etc.).

This checklist ensures the backend and frontend are prepared for full proctoring support.

Phase 4 Implementation Plan



Each task is estimated in developer-days. Core lockdown and monitoring features (sections “Browser & Environment Controls” and parts of “Recording & Snapshot”) should be delivered first. Machine-learning capture and review UI can proceed in parallel. The plan builds in testing and rollout before the end of Phase 4.

Recommended UI Components

- **Home Page:** Clear sections for features (“Authoring & Auto-Marking”, “Secure Proctoring”, “Analytics” etc.), trust badges (ISO/GDPR logos), customer logos/testimonials, and CTAs (Free Trial). Use white background, highlight bars/cards with a warm accent color for primary actions.
- **Unified Auth Page:** A single page with toggles or tabs for “Teacher/Administrator” vs “Student”. Each has a card: Teachers enter email/password; Students enter exam key (and optionally name). Include “Forgot Password” for teachers and exam-key help.
- **Sign In / Registration Cards:** Minimal form fields with labels (Username/Email, Password). For registration, include a dropdown for “Role” (Teacher vs Admin vs Student) if self-registration is allowed. Use rounded input boxes and a prominent Submit button.
- **Exam Room UI (Student):** Central white area with question content. Sidebar (or bottom toolbar on mobile) with navigation between questions, and a chat/help toggle. Include icons/badges for connection status, remaining time, and (if recording) a blinking REC indicator. All controls in orange (or chosen accent).
- **Teacher Monitoring Dashboard:** A table or grid showing each student (name, status, progress bar, flags count). Use color-coded indicators (green/yellow/red) for no flags vs issues. A side panel (or modal) shows a selected student’s live view: current question, time elapsed, and any real-time alerts. Include a chat panel or warning message box.
- **Review/Evidence Panel:** For completed exams, show a timeline slider with paired screen+webcam thumbnail frames (as Synap does【33†L414-L423】). Flag icons on the timeline for detected breaches. Controls to play video or view snapshots at any point. Fields for teacher comments and final decision (Accept/Reject).

These components match modern UX patterns: clean layouts, consistent padding, and visual emphasis on important actions (e.g. “Start Exam” button in orange). Icons and tabs should aid navigation. The interface should look **white, elegant, minimal, and warm**, avoiding unnecessary decoration but using subtle shadows and rounded corners for a premium feel (inspired by the requested “Minimal White + Orange” direction).

Sources: Official docs from Exam.net【6†L49-L51】【9†L83-L89】 and Synap【16†L199-L207】【33†L474-L483】 were used to derive the feature lists, workflows and privacy details. Where direct information was unavailable, notes are marked *unverified*. The implementation timeline and UI suggestions are based on standard SaaS design principles and the analyzed capabilities of the two platforms. The migration checklist is informed by typical Django integration practices and the features identified above.
